

Lab 1

Setup Guide

CS435

Fall 2026

1 Introduction

The purpose of the first lab is to ensure that everyone can access the computing resources needed to run distributed jobs. All students in the course are assigned **8 machines** to act as their “personal cluster.”

Remember that other students in this class as well as those in different classes may be using the same machines as you simultaneously. For that reason, it is important to be mindful of the resources you have. If your jobs are taking a long time to finish, consider optimizing your code as much as possible.

This tutorial will walk you through the steps of configuring Hadoop (HDFS) and Spark on the department machines. While this first lab will not use any HDFS capabilities, it will be required for future labs and programming assignments.

2 Prerequisites

Before starting installation, ensure that these requirements are met:

- You have an active CS account. For those new to the school or department this may not be the case. Send an email to `Compsci_SNA@Mail.colostate.edu` if you are unsure.
- You can use Passwordless SSH¹ by following the instructions from SNA. This means that if you are on a department machine you can login to any other department machine simply using `ssh <hostname>` and do not get prompted for a password.

¹<https://sna.cs.colostate.edu/remote-connection/ssh/keybased/>

3 Instructions

Once your account setup is complete, you can begin installation:

1. Add these lines to the `.bashrc` file in your home directory:

```
source /etc/profile.d/modules.sh module
module purge
module load courses/cs435
```

If you have any other modules which would cause a conflict (particularly any Java or Python versions), those must be commented out. Once this is done, you can either log out and back in again, or **source** the file.

2. You are provided with the necessary configuration files. These have been zipped and can be added to your home directory with the command
`unzip /s/$HOSTNAME/a/nobackup/cs435/$USER/$USER.zip -d ~/`
3. Navigate into the newly created `bigdata_configs` directory and run the `setup_all_config.sh` script located inside.
4. This will generate a new file containing your specific machines available for use. You can read through the entire contents, or just inspect the last part with `tail README.md`
5. Note which machine is listed as the `namenode` and `ssh` into it
6. Run the command `hdfs namenode -format` and confirm when prompted. This allocates your storage space for the next lab.

To ensure that setup completed successfully, run the following commands in this exact order:

```
start-dfs.sh
start-master.sh
start-workers.sh
stop-workers.sh
stop-master.sh
stop-dfs.sh
```

If these all run without errors, you are done! You will use these same commands when working on labs and programming assignments.